

ARTICLE

A numerical model capable of accurately simulating the punching shear behavior of a reinforced concrete slab

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Abstract

The paper presents the tridimensional numeric model of an inner slab-column connection of reinforced concrete, which was previously tested against the punching shear in the lab of the University of Brasília. The numerical analysis has been carried out by the finite element method, using the software ABAQUS. To evaluate the sensitivity of the main factors that influence the responses of the model, a parametric investigation was conducted as per the finite element mesh, mechanical properties of the concrete and plastic parameters of the assumed model for the concrete, named by concrete damaged plasticity model (CDPM). The results of the resistance capacity, displacements, strains, cracking pattern, and failure mode were compared with those obtained experimentally. From this comparison, it was concluded that the numerical model was able to reproduce the punching shear phenomenon with a high level of precision.

KEYWORDS

concrete damaged plasticity, FEM analysis, flat slab, punching shear

1 | INTRODUCTION

Punching shear failure of a reinforced concrete slab-column connection can occur without the yielding of the slab flexural reinforcement and the crushing of the concrete on the compressed face, thus, characterizing an abrupt failure by shear stress. Despite the punching shear being a local failure in the structure, its occurrence causes the redistribution of efforts to a configuration in which the structure was not dimensioned. This may further cause the rupture of other parts of the structure of a building in a kind of chain reaction called progressive collapse.

Punching shear can be associated with relatively recent accidents.^{1–3} However, such a phenomenon has attracted the attention of researchers since the middle of the last century, when investigations were carried out based on experimental tests which allowed the study of punching shear in experimentally monitored results.^{4–8}

In the scope of modern research in structural engineering, the properly calibrated nonlinear finite element models can be very useful to supplement the experimental observations and extend the study through parametric analyses. However, the computational modeling of the punching shear in reinforced concrete flat slabs is not trivial. Aspects such as constitutive models of materials, type of finite elements, and numerical solution procedures, added to the complexity of the stress state present in the punching shear phenomenon, show its complexity.

Discussion on this paper must be submitted within two months of the print publication. The discussion will then be published in print, along with the authors' closure, if any, approximately nine months after the print publication.

Recent studies present the concrete slab punching shear through the finite element method (FEM).^{9–15} From this perspective, observing the current aspects and aiming contribute to the numerical modeling of punching shear in flat reinforced concrete slabs by FEM, this paper aims to propose a numerical model capable of providing results with a high degree of precision compared to experimental results. The experimental and the numerical results are compared to each other in terms of resistance capacity, vertical displacements, concrete strains, tensile reinforcement strains, cracking pattern, and failure mode. Furthermore, a parametric investigation is conducted to assess the sensitivity of such parameters in response to the numerical modeling of punching shear. In this regard, the following parameters have been evaluated: the finite element mesh, tensile strength, modulus of elasticity, and fracture energy of the concrete, and the parameters of the concrete damaged plasticity model (CDPM) adopted for the concrete.

2 | EXPERIMENTAL TEST DESCRIPTION

The experimental test results were used as a reference for the calibration and the validation of the numerical model. This experiment was carried out in the laboratory of structures of the University of Brasília by Palhares¹⁶ and Lima,¹⁷ which is part of a series of essays published by Lima et al.¹⁸ The experimentally tested slab was named RS. Its dimensions in the plan were 2500 × 2500 mm with a thickness of 180 mm. The column was located in the center of the slab and had a square cross-section of 300 × 300 mm and exceeded the upper and lower face of the slab by 800 and 600 mm, respectively.

The slab was reinforced against flexure in both directions with bars measuring 16 mm in diameter spaced 155 mm apart on the tensioned face ($\rho = 0.92\%$). On the compressed face, bars of 10 mm in diameter and with the same spacing as the bars on the tensioned face were used. The concrete cover was equal to 20 mm on both sides. It is noteworthy that shear reinforcements were not used in the slab. On the free edges, “U”-shaped hooks composed of 16 mm bars and 155 mm spacing have been used. The column was reinforced with eight bars of 16 mm in diameter and its stirrups with bars of 10 mm in diameter and 80 mm spacing. Figure 1 shows the details of the slab reinforcements.

The test system consisted of the application of symmetrical loading through eight concentric points concerning the center of the column (Figure 2a). The lower and upper ends of the column were set with the aid of a

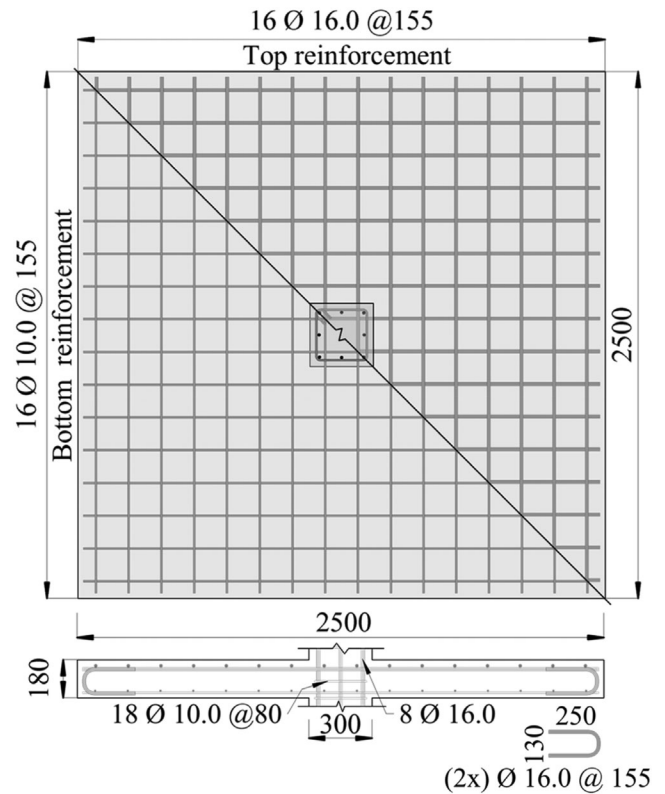


FIGURE 1 Details of reinforcement (measurements in mm)

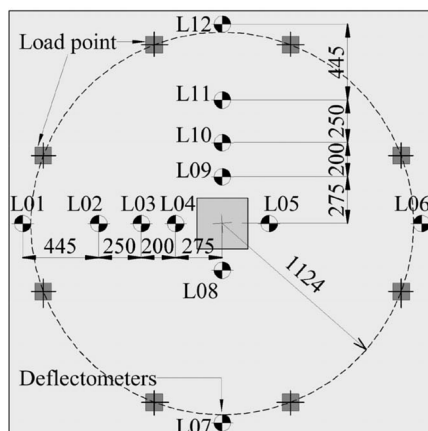
metallic frame and the slab consisted of free edges. To assess the slab's ductility, 12 deflectometers were installed on the upper face of the slab (Figure 2a). Analyzing the possibility of failure due to concrete crushing, four strain gauges were also attached, close to the column, in the radial and tangential directions of the lower face of the slab (Figure 2b). To check for any failure due to flexural of the slab, seven flexural reinforcement bars were instrumented with strain gauges installed at 40 mm from the adjacent face of the column (Figure 2b). Mechanical properties of steel and concrete were extracted from experimental tests and are presented in Table 1.

3 | COMPUTATIONAL MODELING

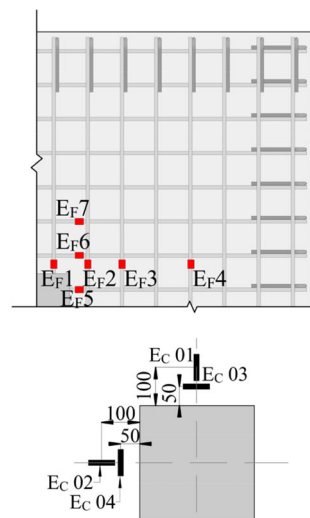
3.1 | Aspects of the numerical model

The numerical model was elaborated in ABAQUS software, which uses the FEM and allows analyses considering physical nonlinearities resulting from geometry and material.¹⁹ As the loading system and structure presented symmetry in both orthogonal axes, it was possible to take advantage of this characteristic and model only on $1/4$ of the experimental slab RS. To make the experimental test system compatible with the reduced model, the

FIGURE 2 RS slab testing and instrumentation system. (a) Loading system and position of deflectometers. (b) Position of strain gauges on steel bars and concrete



(a)



(b)

TABLE 1 Material properties of the tested slab

Concrete	Reinforcement steel			
	Ø 16.0		Ø 10.0	
f_c (MPa)	f_y (MPa)	E_s (GPa)	f_y (MPa)	E_s (GPa)
29.9	549.0	196.9	515.0	126.6

appropriate boundary conditions were applied in the symmetry planes, as shown in Figure 3a.

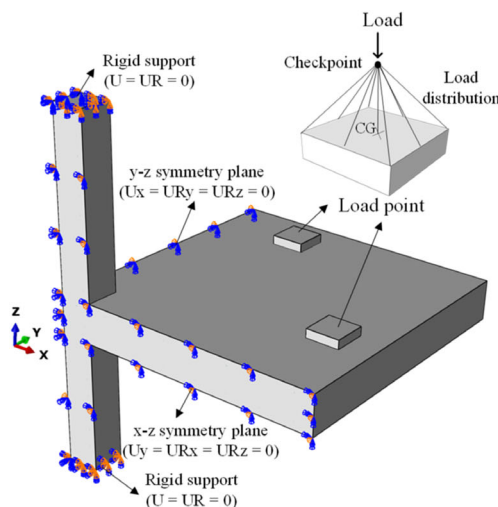
The interaction between the slab and the surface of the loading application plates was defined as a tie type. For concrete and steel reinforcement (Figure 3b), an embedded type connection was employed. Loading was applied to two control points, created on the geometric

center of the load plates. The interaction between the control point (loading) and the top surface of the loading plate was modeled as a kinematic coupling (Figure 3a). The implicit general static load analysis that applies the traditional Newton–Raphson iterative method for model convergence was used, as the load was applied gradually and slowly in the experimental test. Regarding the finite element mesh, as its characteristics can significantly influence the numerical results, a sensitivity study was carried out and is detailed in the next section.

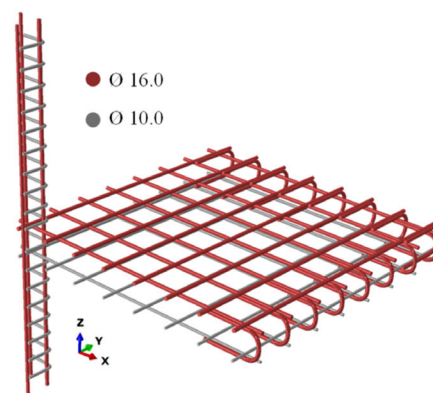
3.2 | Constitutive model of concrete

For the modeling of the nonlinear behavior of concrete, a model was used that combines damage and plasticity,

FIGURE 3 Aspects of the numerical model. (a) Geometry and boundary conditions. (b) Reinforcement arrangement



(a)



(b)

called CDPM, which was implemented in the ABAQUS software library.^{20,21} It was developed specifically for numerical applications of concrete material and assumes that its behavior is characterized by a yielding criterion that depends on the following parameters: the ratio between the initial compressive strengths in the biaxial and uniaxial state (σ_{b0}/σ_{c0}); dilation angle (ψ); eccentricity (ϵ), shape factor (K_c); and viscosity (μ).

The uniaxial tensile behavior of the concrete was represented in two sections (Figure 4), the first being associated with the non-cracking of the concrete, represented by a linear relationship between tensile stress σ_t and strain ϵ_t . This behavior is limited to the maximum tensile strength of concrete f_{tm} and its stiffness is governed by the modulus of elasticity of the concrete E_0 . The modulus of elasticity E_0 was calculated from the equations of *fib* Model Code 2010.²³ The maximum tensile strength f_{tm} was calculated as a function of the compressive strength of concrete f_{cm} , as proposed by Genikomsou and Polak.¹⁰ The second section of the tension behavior, known as softening, is associated with the beginning of the cracking of the concrete until the maximum crack opening W_c , in which the rupture of the material is reached with the annulment of the tension in it. The equating of the softening section was assumed according to Hordjik.²⁴ The fracture energy was estimated using the equation recommended by the *fib* Model Code 2010. To reduce the dependence of the results on the mesh discretization, Alfarah et al.²² mention that it is desirable not to work with the concept of crack opening, but with the concept of cracking strain, which considers a single crack per element. This hypothesis is suitable to deal with the global behavior of the structures in numerical simulations, also being used by Genikomsou and Polak.¹⁰ The transformation of the crack width variable W to cracking strain can be performed through the division W/l_{eq} , where l_{eq} is the length characteristic of the finite element used for concrete discretization. Alfarah et al.²² define the l_{eq} as its

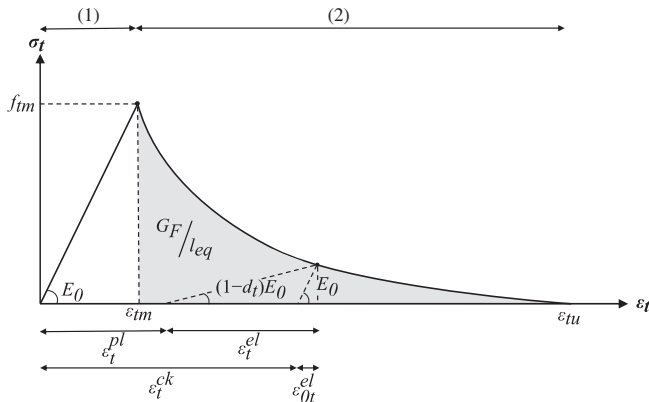


FIGURE 4 Uniaxial concrete tensile behavior²²

volume divided by the area of its largest face. Therefore, the total strain of the concrete, in the uniaxial tensile state can be calculated by the sum of the strain associated with the peak tensile stress ϵ_{tm} plus the cracking strain.

The uniaxial compressive behavior of concrete followed the normative recommendations of the *fib* Model Code 2010²³ in the hardening of the material and its softening was modeled as proposed by Krätzig and Pölling.²⁵ This concrete model was implemented by Alfarah et al.²² and is represented in three sections (Figure 5), so that the first is limited to 40% of the concrete compressive strength, following the linear elastic behavior. The second section is called hardening, as the concrete starts the cracking process accompanied by loss of stiffness, although it still manages to absorb load until reaching maximum compression. According to *fib* Model Code 2010, this segment is quadratic, in which E_{ci} is the initial modulus of elasticity, f_{cm} is the compressive strength of concrete, ϵ_{cm} is the specific strain at the stress peak. f_{cm} was adopted as 29.9 MPa, as obtained experimentally, and ϵ_{cm} can be assumed to be 0.0022 for low strength concrete grades. The third section of concrete softening, proposed by Krätzig and Pölling,²⁵ is calculated iteratively and depends on the crushing energy per unit area G_{ch} , the l_{eq} and the relationship between the plastic crushing strain ϵ_c^{pl} and the inelastic crushing strain ϵ_c^{ch} . The value of the crushing energy G_{ch} (N/mm) was estimated as proposed by Oller.²⁶ Another variable that can be highlighted is the localized crushing energy, given by the G_{ch}/l_{eq} ratio. This can be represented mathematically by the area under the stress-strain curve of Sections 2 and 3, so that the ultimate strain ϵ_{cu} is found when such equality is satisfied (Figure 5).

The calculation of the damage variables was performed according to Alfarah et al.²² The evolution law proposed by these authors considers the damage variables, associated with a given plastic strain, as a portion of the normalized energy dissipated by the damage.

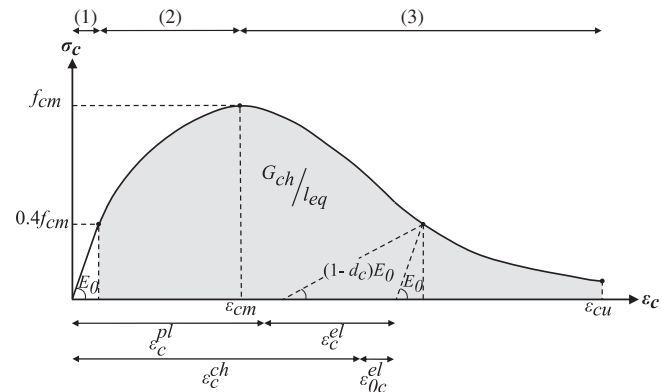


FIGURE 5 Uniaxial concrete compression behavior²²

TABLE 2 Summary of the equations of the constitutive model of concrete

	Tensile behavior	Compressive behavior
Section 1	$\sigma_{t(1)} = E_0 \cdot \varepsilon_t$ where: $E_0 = \alpha_i \cdot E_{ci}; E_{ci} = 10,000 \cdot \alpha_E \cdot (f_{cm})^{1/2};$ $\alpha_i = 0.8 + 0.2 \cdot \frac{f_{cm}}{88} \leq 1.0$	$\sigma_{c(1)} = E_0 \cdot \varepsilon_c$
Section 2	$\sigma_{t(2)} = f_{tm} \cdot \left\{ \left[1 + \left(3 \cdot \frac{W}{W_c} \right)^3 \right] \cdot e^{-6.93 \frac{W}{W_c}} - \frac{W}{W_c} \cdot 28 \cdot e^{-6.93} \right\}$ where: $f_{tm} = 0.33 \cdot \sqrt{f_{cm}}; W_c = \frac{5.14 \cdot G_F}{f_{tm}}; G_F = 0.073 \cdot f_{cm}^{0.18};$ $\varepsilon_t = \varepsilon_{tm} + \frac{W}{W_{eq}}$	$\sigma_{c(2)} = \frac{E_{ci} \cdot \frac{f_{cm}}{f_{cm}} - \left(\frac{\varepsilon_c}{\varepsilon_{cm}} \right)^2}{1 + \left(E_{ci} \cdot \frac{\varepsilon_{cm}}{f_{cm}} - 2 \right) \cdot \frac{\varepsilon_c}{\varepsilon_{cm}}} \cdot f_{cm}$
Section 3	Not applicable	$\sigma_{c(3)} = \left(\frac{2 + \gamma_c \cdot \frac{f_{cm}}{f_{cm}} \cdot \varepsilon_{cm}}{2 \cdot f_{cm}} - \gamma_c \cdot \varepsilon_c + \frac{\varepsilon_c^2 \cdot \gamma_c}{2 \cdot \varepsilon_{cm}} \right)^{-1}$ where: $\gamma_c = \frac{\pi^2 \cdot f_{cm} \cdot \varepsilon_{cm}}{2 \cdot \left[\frac{G_{ch}}{f_{eq}} - 0.5 \cdot f_{cm} \cdot \left(\varepsilon_{cm} \cdot (1-b) + b \cdot \frac{f_{cm}}{E_0} \right) \right]^2};$ $b = \frac{\varepsilon_c^{pl}}{\varepsilon_c^{ch}}; G_{ch} = \left(\frac{f_{cm}}{f_{tm}} \right)^2 \cdot G_F$
Damage	$d_t = 1 - \frac{1}{2} \left[4 \cdot e^{-b_t \cdot \varepsilon_t^{c_k}} - e^{-2b_t \cdot \varepsilon_t^{c_k}} \right]$ where: $b_t = \frac{f_{t0} \cdot l_{eq}}{G_F} \cdot \left(\frac{3}{2} \right)$	$d_c = 1 - \frac{1}{9.873} \cdot \left[17.746 \cdot e^{-b_c \cdot \varepsilon_c^{ch}} - 7.873 \cdot e^{-2b_c \cdot \varepsilon_c^{ch}} \right]$ where: $b_c = \frac{f_{c0} \cdot l_{eq}}{G_{ch}} \cdot \left(1 + \frac{7.873}{2} \right)$

Table 2 summarizes the equations that describe the behavior of concrete under uniaxial stress described in this section and schematically in Figures 4 and 5. The Poisson's ratio was assumed as equal to 0.2 for concrete, as suggested by *fib* Model Code 2010.²³

3.3 | Constitutive model of steel

As punching shear failure is mainly related to concrete failure, so that the studied slab-column connection has been designed in a way that there is no excessive yielding of the flexural reinforcement, the perfectly elasto-plastic model was considered. The steel properties used were the same as those obtained experimentally and presented in Table 1. The value of the Poisson's ratio was assumed to be 0.3.

4 | PARAMETRIC RESEARCH AND MODEL CALIBRATION

This section will emphasize analyzes related to mesh sensitivity, plastic parameters of CDPM, and mechanical properties of the concrete. In those analyzes, when a parameter is evaluated, all the other parameters are considered with their calibrated values, which are presented throughout this section and are listed in Tables 3 and 4.

To evaluate the accuracy level of the numerical model, it will be compared to the experimental slab

TABLE 3 Plastic parameters of the CDPM

ψ	ϵ	σ_{b0}/σ_{c0}	K_c	μ
25°	0.1	1.16	0.667	0.0001

through the load–displacement relationship as well as the observation of the cracking pattern and formation of the punching shear cone in the numerical analyses. It is noteworthy that the displacement was considered as the mean of the values obtained by the deflectometers furthest from the column, L01, L06, L07, and L12, in the experimental slab and the respective deflectometers in the numerical model, since they are the most sensitive to deflection and the slab had concentric loading.

To calibrate and determine the ultimate load of the punching shear resistance, analyzes were performed by imposing displacement on the loading plates, because, at the peak load, a fragile behavior is expected, with a pronounced drop in load in the load–displacement graph. On the other hand, Newton Raphson's method does not admit decreasing values of the control variable, so that load imposition would not allow the identification of the ultimate load. Next, the comparison between the numerical and experimental results was made considering the imposition of load in the numerical model obtained after calibration, since the RS slab test was performed by load control. This analysis strategy was also used by Sanabria et al.¹²

TABLE 4 Concrete modeling parameters

Concrete parameters	Value	Reference
Mean compression strength	29.9 MPa	Test
Mean tensile strength	1.80 MPa	Genikomsou and Polak ¹⁰
Modulus of elasticity	21.551 MPa	<i>fib</i> Model Code 2010 ²³
Poisson's ratio	0.2	<i>fib</i> Model Code 2010 ²³
Fracture energy	0.134 N/mm	<i>fib</i> Model Code 2010 ²³
Crushing energy	39.97 N/mm	Oller ²⁶
Characteristic length	30 mm	Alfarah et al. ²²
Tensile damage	–	Alfarah et al. ²²
Compression damage	–	Alfarah et al. ²²
Tensile softening	–	Hordjik ²⁴
Compression softening	–	Krätzig and Pölling ²⁵

4.1 | Mesh sensitivity

For continuous linear elements, Wosatko et al.,¹³ recommend the use of at least 10 elements in the direction of the thickness of the slab for representing the mode of rupture by punching shear. As for the quadratic elements, which are more refined, Hendriks et al.²⁷ suggest at least six elements in the direction of the thickness of the slab for the evaluation of complex tension states. The recommendations proposed by these authors were assumed to evaluate the elements C3D8, C3D8R, C3D20 and C3D20R, available in the ABAQUS library, in a way that four different models have been processed, composed by distinct mesh (Table 5). It is worth pointing out that the methodology applied in the modeling of concrete, according to Alfarah et al.,²² allows for us to minimize the sensitivity of the numeral results in relation to the size of the assumed element, favoring the comparison of the results of Table 5 only in terms of the type of element.

TABLE 5 Element type analysis

Mesh	Element type	Element size (mm)	Number of elements in slab thickness
A	C3D8	18	10
B	C3D8R	18	10
C	C3D20	30	6
D	C3D20R	30	6

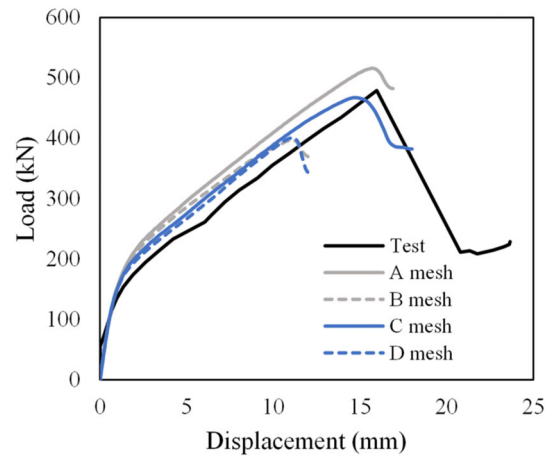


FIGURE 6 Load-displacement graph of meshes A, B, C, and D

The load-displacement curves of the numerical slabs composed by different types of elements and of the experimental slab are presented in Figure 6. Whereas, Figure 7 illustrates the cracking pattern of the numerical models, from the cross-section close to one of the column's faces in a post-rupture instant.

From Figure 6, it is possible to notice that the C3D8 element presents a slightly stiffer behavior than the others and a peak load higher than the one obtained in the experimental test. Characteristics such as underestimation of deflection in slabs and increased strength of the structure occur due to the effect known as shear locking, typical of linear elements of complete integration, such as the C3D8.²⁸ On the other hand, the C3D8 element formed a punching shear cone with good post-rupture precision. The mesh formed by the C3D8R elements showed lower stiffness when compared to the mesh of C3D8 type elements, demonstrating that the reduced integration is effective in combating the undesirable effect of shear locking, although it is still possible to identify a stiffness that is higher than the experimental test due to the locking generated by the linear interpolation between the nodes, which are incompatible with the curvatures produced by the bending of the slab. Further on, it is observed that the mesh of C3D8R elements has a resistance capacity substantially lower than the peak load obtained in the experimental test. Although the models

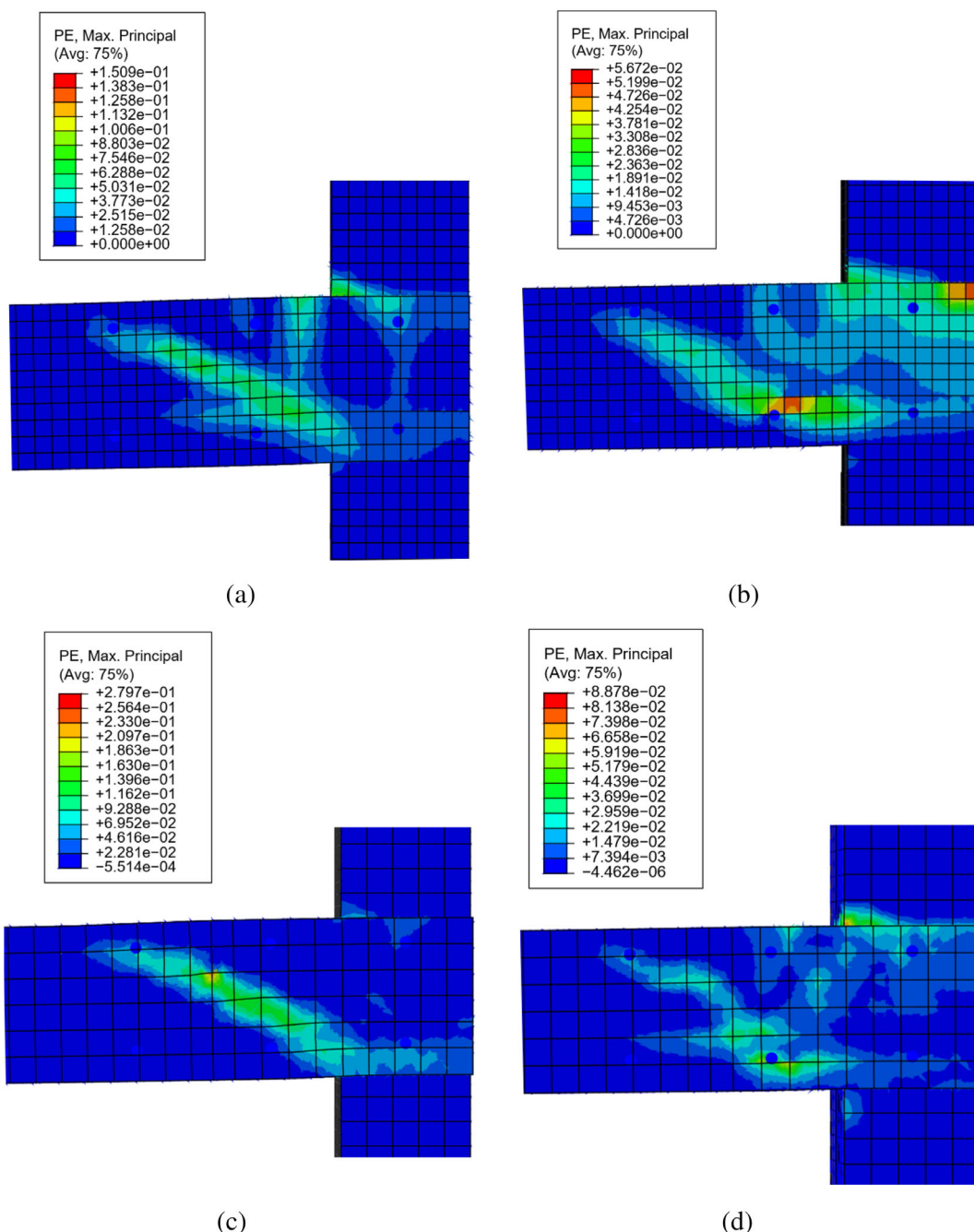


FIGURE 7 Crack pattern of slabs with different types of elements. (a) C3D8; (b) C3D8R; (c) C3D20; (d) C3D20R

have been processed with hourglassing control, the low resistance capacity is possibly associated with some level of this effect, a hypothesis that is strengthened by the instability of the mesh that provided difficulty in the convergence of the model even with the use of fewer integration points when compared to the C3D8 element, in addition to not forming the punching shear cone in the analysis of the crack pattern in the post-rupture, producing a rupture mode not coherent with the one observed experimentally (Figure 7). Still in Figure 6, it can be seen that the result obtained with the C3D20R mesh of

elements is the least stiff of all, in addition to being the closest to the stiffness of the experimental slab. However, instabilities in the mesh produced a premature rupture of the slab, not reaching the ductility present in the experimental test, nor even typical cracking patterns of punching shear cone formation, configuring a behavior similar to that which occurred with the C3D8R element. With all the types of tested elements, C3D20 presented a behavior more coherent with what was observed experimentally, both in terms of resistance capacity and in the formation of the punching shear cone post-rupture

TABLE 6 Element size analysis

Mesh	Element type	Element size (mm)	Number of elements in slab thickness
E	C3D20	45	4
F	C3D20	36	5
C	C3D20	30	6
G	C3D20	25.7	7

(Figure 7). Therefore, this one will be used in subsequent analyzes following the calibration.

Next, a sensitivity study for different sizes of the mesh elements was conducted in such a way that four possibilities were tested, varying the number of elements in the direction of the thickness of the slab, as shown in Table 6. Figure 8 shows the comparison between the behavior of meshes of different sizes and the experimental slab through the load–displacement graph.

From Figure 8, it is observed that all models have practically the same rigidity, which indicates that the methodology proposed by Alfarah et al.²² is effective in minimizing variations in the numerical results in relation to the mesh. However, the thicker meshes (E and F) underestimated the slab's strength capacity and did not show a drop in load at the time of failure, not characterizing the fragile failure mode, typical of the punching shear phenomenon. Therefore, to model slabs subject to punching shear, a local rupture phenomenon with stress concentration, it is necessary to refine the mesh in the region of the slab-column connection. As for the cracking pattern, only the E mesh did not form inclined cracks on the rupture surface, the other meshes managed to reproduce the punching shear cone. The more refined mesh (mesh G) presented a behavior very similar to that obtained with mesh C, indicating the convergence of results, but with a drastic increase in computational cost (about three times the time required for mesh C). Considering the results obtained for the different sizes of meshes studied, it is concluded that the use of mesh C was the most viable for presenting the best computational precision-cost ratio.

Still, in an attempt to reduce the computational cost of the model, an alternative mesh named H was tested, prioritizing the region of the slab-column connection (Figure 9). The H mesh was able to reduce the computational cost without losing precision in relation to the C mesh (Figure 10), so the H mesh was used in the subsequent analyses.

In the discretization of the steel reinforcement, linear elements of truss type T3D3 were used. It features three nodes, each having three degrees of translational freedom and quadratic interpolation order. In the case of a perfect bond between steel and concrete, Hendriks et al.²⁷

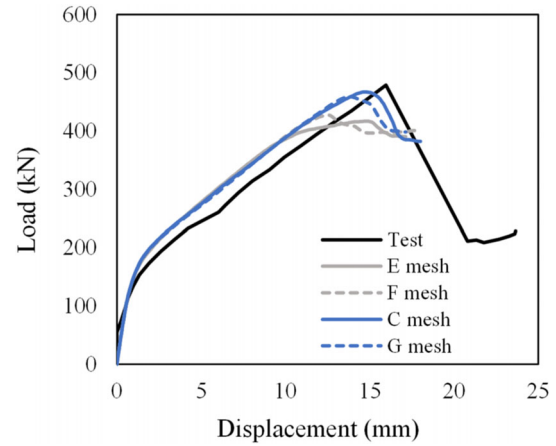


FIGURE 8 Load–displacement graph of meshes E, F, C, and G

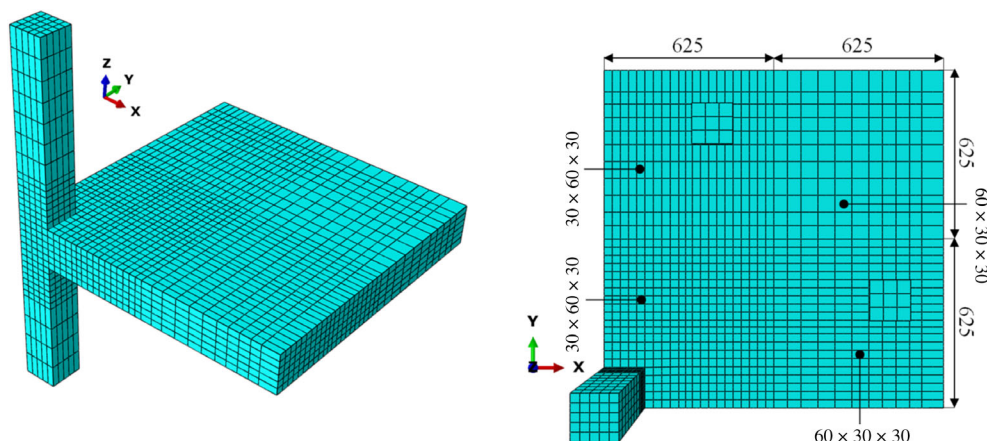
recommend that the interpolation order of the reinforcements is the same as the one adopted by concrete. This consideration promotes better mesh compatibility, avoiding divergent strain modes between the materials. The finite element size of the reinforcement was set at 15 mm, which corresponds to half of the characteristic length of concrete. No detailed calibration was carried out in relation to steel reinforcements, as the failure is expected to occur in concrete. Finally, the model was discretized by 40,122 nodes, 6787 elements of type C3D20 and 3810 elements of type T3D3.

4.2 | Investigation of CDPM plastic parameters

One of the most important parameters in the calibration of the CDPM is the dilatancy angle (ψ), not only because of the sensitivity that the model presents in relation to this variable, but also due to the divergence found in literature about it. Vermeer and Borst²⁹ suggest a value of $\psi = 13^\circ$, which should not be higher than 20° . Other authors suggest higher values for the angle of dilatancy, such as Voyiadjis and Taqieddin³⁰ who propose an amplitude between 31° and 42° , against the standard ABAQUS value of $\psi = 36^\circ$. In this sense, a wide range of dilatancy angle values was analyzed, ranging from 15° to 40° , as

FIGURE 9 H mesh details.

(a) Perspective view. (b) Top view (units in mm)



(a)

(b)

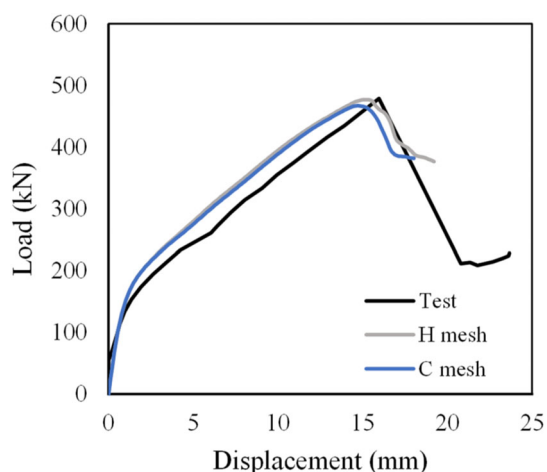


FIGURE 10 Load-displacement graph of meshes H and C

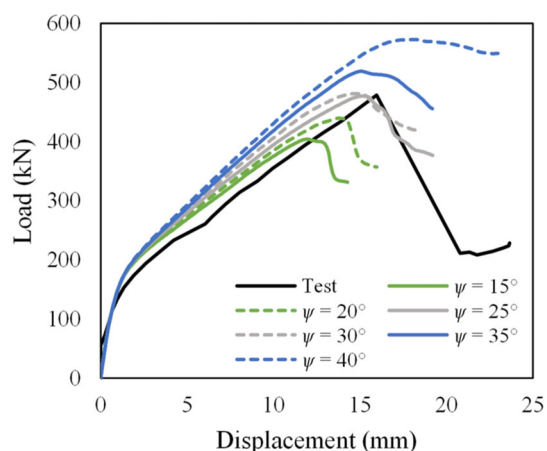


FIGURE 11 Load-displacement graph for different angles of dilatancy

shown in Figure 11. It is observed that the greater the dilatancy angle, the greater the stiffness, ductility, and load capacity of the numerical model. Another

interesting aspect is that the exaggerated increase in the dilatancy angle can modify the failure mode of the concrete, as shown in Figure 11, in a way that, for $\psi = 40^\circ$, the load-displacement graph does not show a sudden load drop, not showing the phenomenon of punching shear and tending to a rupture by flexural-punching shear, since there was a significant flow of the yielding of flexural reinforcement along with the formation of the punching shear cone, going against what was experimentally obtained.

All processed models present stiffness higher than the one obtained in the experimental one. However, the lower the value of ψ , the more the stiffness of the numerical model resembles the experimental one. This pattern of behavior was reported by Wosatko et al.,¹³ in a way that even the lowest values of the dilatancy angle such as $\psi = 5^\circ$ promote an overestimation of the volumetric expansion effect in concrete for low and intermediate loading stages, resulting in excessive stiffness of the numerical model. This is a limitation of CDPM. Despite this, the use of $\psi = 25^\circ$ produces volumetric variations similar to those obtained experimentally for high loading stages, showing itself capable of reproducing concrete failure modes with reasonable accuracy. Finally, the value of $\psi = 25^\circ$ was chosen for the following analyses, as it better estimated the load capacity of the experimental slab and its ductility (Figure 11), being more appropriate for studies related to punching shear failure. Wosatko et al.¹³ also concluded that the punching shear failure mode is better represented using $\psi = 25^\circ$.

Regarding viscosity (μ), values between 5×10^{-5} and 10^{-3} were tested (Figure 12). Michał and Andrzej³¹ noticed that, for zero viscosity, the cracking of concrete subjected to tension occurs in an extremely localized way, not allowing the propagation of cracks. As for non-zero viscosity values, there is greater crack propagation,

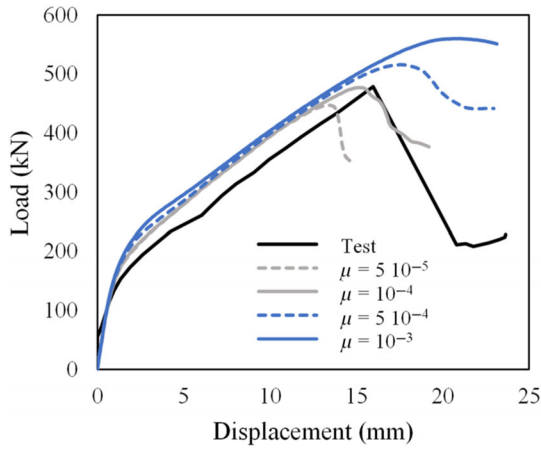


FIGURE 12 Load-displacement graph for different viscosity values

suggesting tension relief (material relaxation) in the most stressed points of the structure and, consequently, greater ductility and load capacity.

From the load-displacement graph (Figure 12), it can be seen that the value of $\mu = 5 \times 10^{-5}$ resulted in low ductility and load capacity, in addition to an excessively fragile rupture, observed by the post-rupture with a very accentuated load drop compared to the one obtained experimentally. The value of $\mu = 10^{-4}$ produced the results that best fit the RS slab, in terms of load capacity, ductility and post-rupture. Therefore, this was used in subsequent analyses. Relatively high viscosity values, such as $\mu = 5 \times 10^{-4}$ and $\mu = 10^{-3}$, allow the excessive propagation of cracks, overestimating the ductility and load capacity of the slab, also increasing stiffness caused by the artificial addition of the linear-elastic section in the concrete behavior.

For the other parameters, the default values of ABAQUS were assumed. As for these values, there is a great consensus in the literature.

4.3 | Mechanical properties of concrete

4.3.1 | Mean tensile strength

Three different equations were tested to calculate f_{tm} , the one recommended by *fib* Model Code (2010),²³ the one used by Genikomsou and Polak¹⁰ and the lower limit of the mean tensile strength, also according to *fib* Model Code 2010 (Table 7).

The load-displacement graph in Figure 13 shows that the f_{tm} value recommended by *fib* Model Code 2010 tends to overestimate the value of the mean tensile strength of concrete, as it produces an excess of load capacity in

TABLE 7 Mean tensile strength of concrete tested in numerical calibration

Reference	Calculation expression	f_{tm} value (MPa)
Lower limit of <i>fib</i> Model Code 2010 ²³	$f_{tm} = 0.21 \cdot (f_{cm} - 8)^{2/3}$	1.65
Genikomsou and Polak ¹⁰	$f_{tm} = 0.33 \cdot \sqrt{f_{cm}}$	1.80
<i>fib</i> Model Code 2010 ²³	$f_{tm} = 0.30 \cdot (f_{cm} - 8)^{2/3}$	2.36

relation to the experimental slab. On the other hand, the normative lower limit results in an excessively conservative premise, as it underestimates the slab's load capacity and its ductility. Genikomsou and Polak¹⁰ also studied the phenomenon of punching shear in flat slabs and the value of the mean tensile strength, according to these authors, produced the best fit results to the experimental slab. Therefore, the value of $f_{tm} = 1.80$ MPa was adopted in the following analyses.

4.3.2 | Modulus of elasticity

In the calculation model of the *fib* Model Code 2010,²³ the influence of the mineralogical nature of the aggregate on the concrete's modulus of elasticity is estimated by the qualitative factor α_E . It was decided to evaluate the behavior of the numerical slab for different values of this factor. The value of $\alpha_E = 1.0$ was set as a reference value and variations were analyzed with a reduction and increase of 20%, corresponding to $\alpha_E = 0.8$ and $\alpha_E = 1.2$,

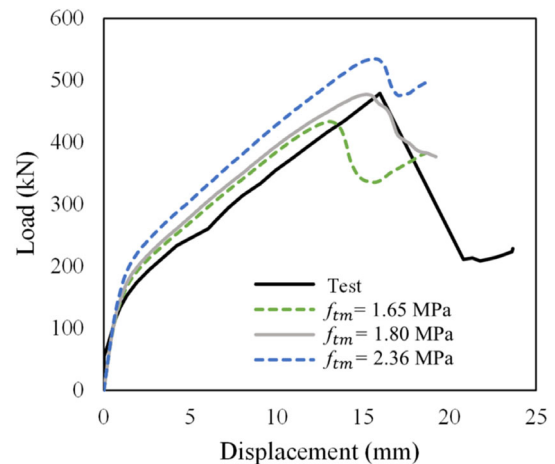


FIGURE 13 Load-displacement graph for different values of mean concrete tensile strength

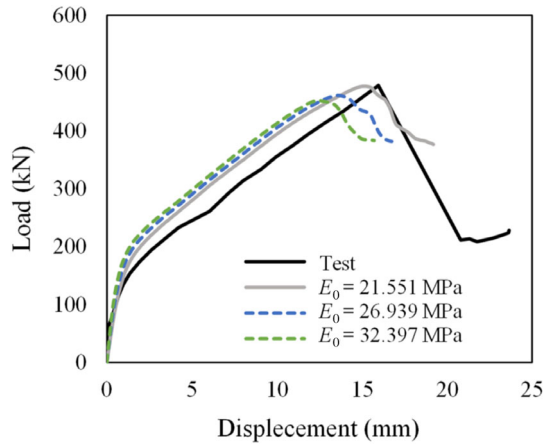


FIGURE 14 Load–displacement graph for different values of concrete modulus of elasticity

respectively. The calculated values of the modulus of elasticity are shown in Figure 14.

From the load–displacement graph (Figure 14), it can be seen that the higher the elastic modulus of the concrete, the stiffer the structure becomes; however, it reduces its ductility and load capacity. On the other hand, a lower modulus of elasticity is related to greater deformability of the numerical slab, capable of increasing its ductility and load capacity. Similar conclusions to these are also reported by Sanabria et al.¹²

The smallest modulus of elasticity of $E_0 = 21,551$ MPa, corresponding to $\alpha_E = 0.8$, provided the results that came closest to the experimental slab, both in terms of stiffness, load capacity, and ductility. Therefore, this was used in the following analyses.

4.3.3 | Fracture energy

Fracture energy (G_F) is a parameter that directly influences the tensile behavior of concrete, from the beginning of cracking until failure (softening). Therefore, a parametric study was carried out in which the value obtained according to the *fib* Model Code 2010 was taken as a reference and variations of 20% both for more and for less were evaluated (Figure 15).

From the graphs in Figure 15, it can be seen that the higher the G_F value, the greater the slab's stiffness, ductility, and load capacity. For the lowest G_F value, the numerical slab underestimated the load capacity of the RS slab and its ductility. Among the estimated values, the G_F reference value, calculated according to the *fib* Model Code 2010, generated the most consistent results with the experimental ones and will be used in the following analyses. Although the crushing energy (G_{ch}) has not been directly considered in the calibration of the numerical

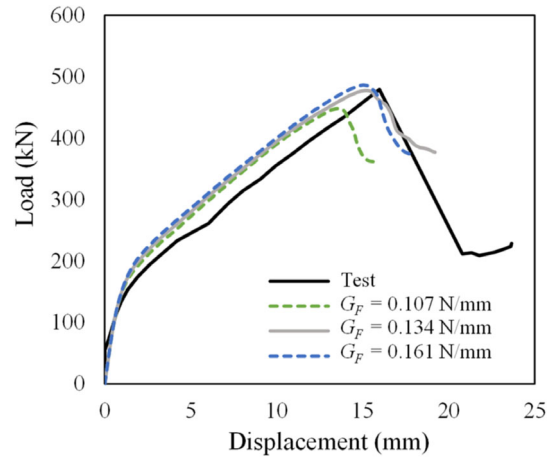


FIGURE 15 Load–displacement graph for different values of concrete fracture energy

model, this variable is calculated as a function of G_F . Thus, changes in G_F cause proportional alterations in G_{ch} .

5 | COMPARISON OF NUMERICAL AND EXPERIMENTAL RESULTS

5.1 | Ultimate vertical loads and displacements

After determining the peak load by displacement imposition, the model with load imposition was processed and it was found that both methods had the same behavior. Table 8 presents the comparison of the results of the peak loads (V_u) and the respective vertical displacements (δ_u) obtained experimentally and through the numerical model by load imposition. From these results, it is clear that the numerical model was able to estimate the ultimate load and deflection of the slab with high accuracy.

5.2 | Vertical displacements

In order to evaluate the general slab behavior, in terms of vertical displacements, it was considered the mean value of the deflectometers equidistant to the geometric center of the column. Figure 16 compares the numerical and experimental mean values, in a way that the experimental results are denoted with the suffix “Test” and the numerical ones with the suffix “FEM,” allowing a more thorough analysis of the response of the slab to the load increase. From this one, it can be seen that the numerical results are slightly stiffer, as mentioned previously, and when the numerical slab approaches the ultimate load,

TABLE 8 Comparison of numerical and experimental results

Test slab		Numerical slab		Relative error (%)	
$V_{u,Test}$ (kN)	$\delta_{u,Test}$ (mm)	$V_{u,FEM}$ (kN)	$\delta_{u,FEM}$ (mm)	$\frac{V_{u,FEM} - V_{u,Test}}{V_{u,Test}}$	$\frac{\delta_{u,FEM} - \delta_{u,Test}}{\delta_{u,Test}}$
478.8	15.97	477.1	15.73	-0.35	-1.5

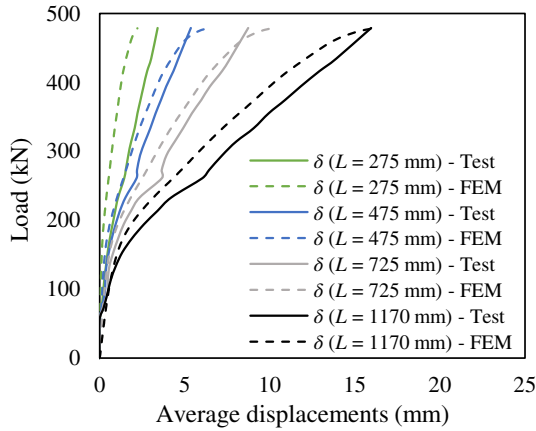


FIGURE 16 Experimental and numerical mean vertical displacements

there is a significant increase in its displacement rate. This increase is justified by the processing of the model through the imposition of load by the Newton Raphson method, so that, when approaching the failure, the model has difficulties in convergence, inducing excessive displacements.

5.3 | Concrete surface strain

To evaluate the model's ability to estimate concrete strains, mean values of tangential and radial strains obtained from the numerical model were compared with the experimental results. The mean of the strain gauges

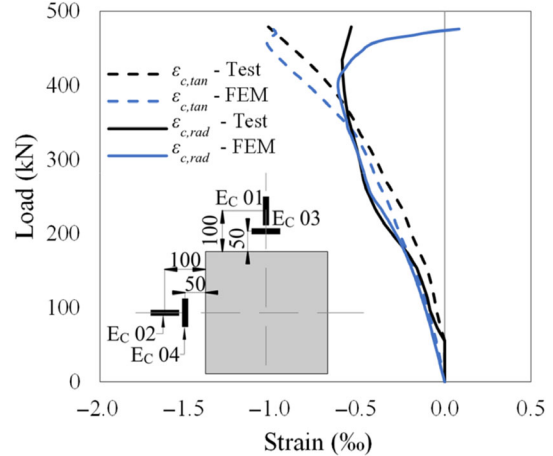


FIGURE 17 Concrete load-strain graph, experimental and numerical strain gauges

EC01 and EC02 gives the radial strains ($\epsilon_{c,rad}$) and one of the strain gauges EC03 and EC04 gives the tangential strains ($\epsilon_{c,tan}$), as shown in the legend of Figure 17. Again, the results denoted with “Test” refer to the experimental and those denoted with “FEM” to the numerical model.

From Figure 17, analyzing the tangential strains, it can be seen that the behavior of the numerical model is very similar to the experimental one, however, it shows slightly less stiff results. The numerical radial strains presented an excellent adjustment to the values obtained experimentally, except in the stretch that precedes the rupture, where there was a more abrupt decompression

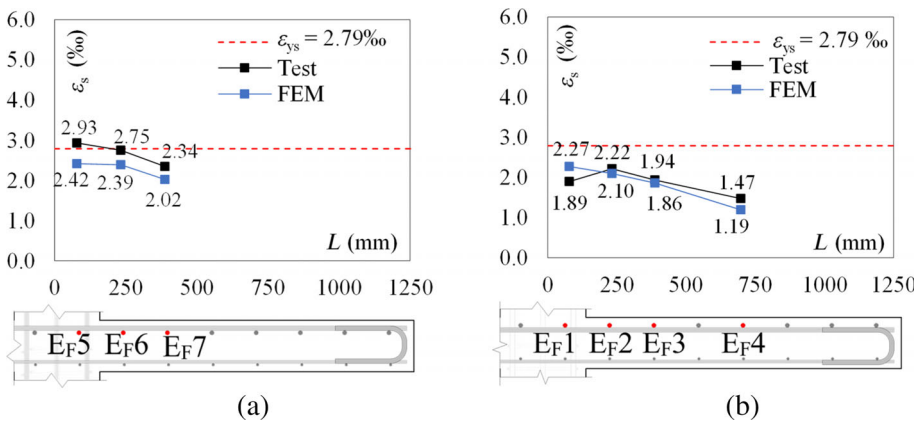
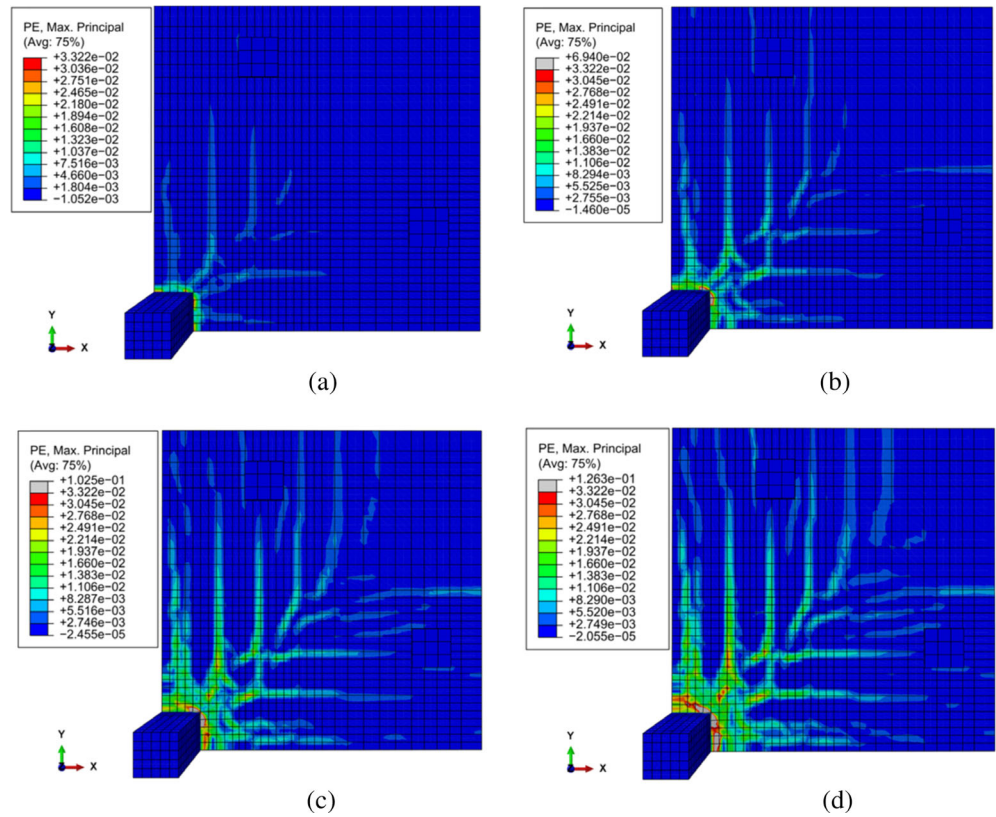


FIGURE 18 Strains in experimental and numerical flexural reinforcement. (a) Direction x; (b) Direction y

FIGURE 19 Crack pattern of the numerical slab.

(a) $0.50 \cdot V_{u,FEM}$; (b) $0.70 \cdot V_{u,FEM}$;
(c) $0.90 \cdot V_{u,FEM}$; (d) $1.00 \cdot V_{u,FEM}$



in the numerical model. This behavior is typical of punching shear failures.^{32,33} Regarding the peak value of the radial strain in concrete, the numerical value found has a magnitude very close to that observed experimentally (approximately -0.60%), however, the numerical result reaches the maximum strain at 84% of the peak numerical load against 90% of the peak load in the case of the experimental one. Finally, as the maximum strain value of concrete (tangential strains of approximately -1.03%) is much lower, in modulus, than the concrete crushing strain value of 3.5% , it can be stated that there was no concrete failure by crushing.

5.4 | Strain of the flexural reinforcements

The behavior of the flexural reinforcements of the numerical model was evaluated through the strain profile in a load stage at the imminence of failure (Figure 18).

From Figure 18, it can be observed that the numerical strain gauges located in the x-direction presented a slightly stiffer behavior than those obtained experimentally. On the other hand, the numerical strain gauges located in the y-direction obtained an excellent fit with the experimental results, except for the E_{F1} which in the experimental test did not follow the expected evolution of

strain in the vicinity of the column and is indicated by the numerical model. Regarding the yield of steel bars, the experimental results indicate the yield of steel at the E_{F5} position and steel strain close to the yield value at the E_{F6} position. The other strain gauges showed strain significantly below the steel yield strain (ϵ_{ys}). The numerical results do not show any flexural reinforcement yielded at the monitoring points, reinforcing the hypothesis that the slab has ruptured by punching shear.

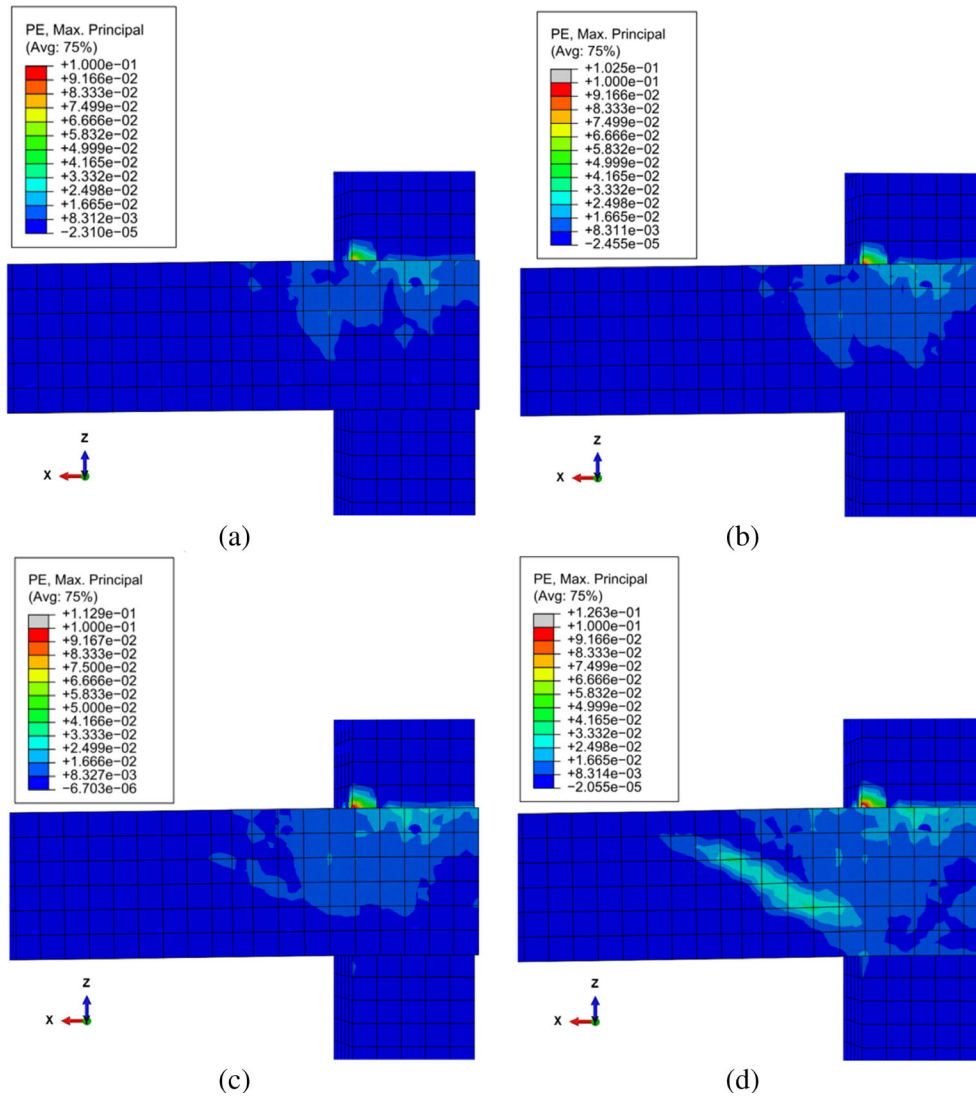
5.5 | Crack pattern and rupture mode

The crack pattern is studied from the top view of the numerical slab for different loading stages. Figure 19 represents the plastic strains of the structure, thus, allowing the identification of cracking patterns and their evolution throughout the simulation.

For 50% of the peak load, the cracking in the numerical slab is relatively low with the first cracks that appeared on the column perimeter, followed by radial cracks. With the increase in load to $0.70 \cdot V_{u,FEM}$, there is an intensification of the radial cracks, which propagate from the column region to the edges, while it is also possible to identify the beginning of cracking tangentially. When 90% of the peak load is reached, the propagation of the radial cracks continues. However, at a lower intensity

FIGURE 20 Analysis of the numerical slab failure mode.

(a) $0.85 \cdot V_{u,FEM}$; (b) $0.90 \cdot V_{u,FEM}$;
(c) $0.95 \cdot V_{u,FEM}$; (d) $1.00 \cdot V_{u,FEM}$



level, tangential cracks begin to develop rapidly. The crack map that precedes the rupture of the numerical slab is marked by a certain stabilization in radial strains and the development of tangential strains around the column. This crack propagation was also observed during the experimental testing of the RS slab and is reported in experimental observations by other researchers.^{34–36}

For a better understanding of how the failure occurred in the computer simulation, the slab cross-section was analyzed at four load levels, ranging from 85% to 100% of the numerical peak load (Figure 20). At lower load levels, up to 90% of the peak numerical load, it is clear that the severely cracked region is located close to the column and on the tensioned face, mainly due to the flexure effort. When a load of 95% of $V_{u,FEM}$ is reached, the combination of shear and flexure forces promote a change in internal forces, so that the magnitude of the inclined cracks begins to prevail. At the peak load, there is a sudden change in the crack pattern, with

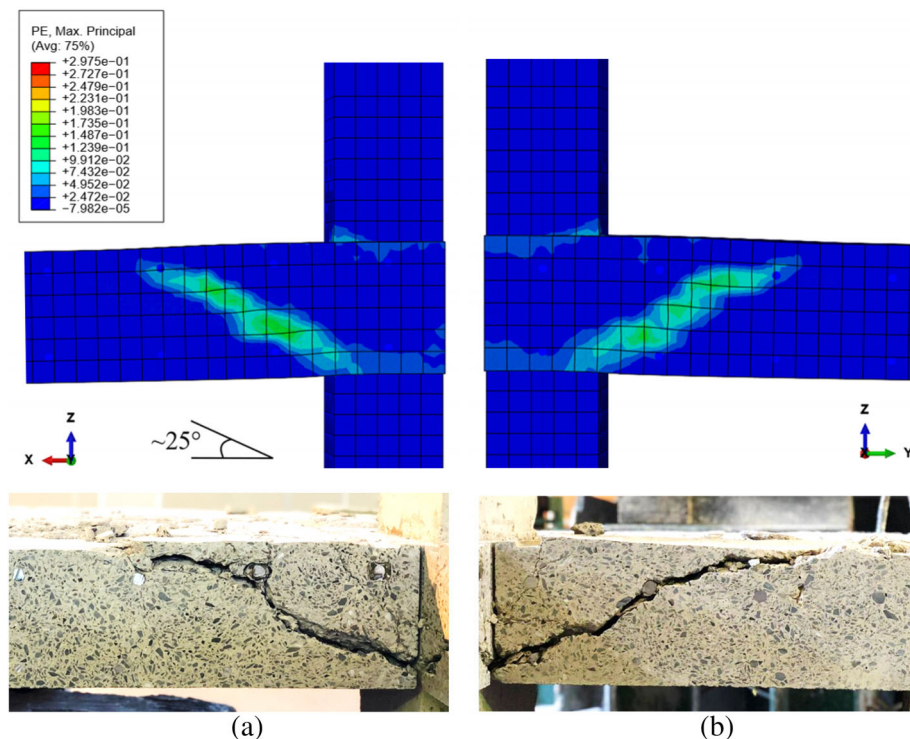
the appearance of the main inclined crack appearing that determines the typical conical rupture surface of punching shear. The sudden appearance of the main inclined crack is common in punching shear collapses, as this behavior demonstrates a fragile failure of the structure.

To investigate the actual failure surface, the experimental slab was cut after the test, allowing a direct comparison of failure mode between the numerical and experimental slab (Figure 21).

From Figure 21, it can be seen that the numerical model and the experimental test have similar rupture surfaces, both in terms of inclination and configuration. The inclination of the rupture surface is approximately 25° in relation to the plane of the slab, in accordance with Regan,³⁷ who concluded that the rupture surface forms an angle of approximately 25° and with CEB-FIP Model Code 1990,³⁸ which points out that the rupture surface is at an angle between 25° and 30° .

FIGURE 21 Numerical and experimental slab failure surface.

(a) Direction x; (b) Direction y



6 | CONCLUSIONS

This paper presented a numerical model capable of simulating the punching shear behavior of a reinforced concrete slab, in terms of resistance capacity, displacements, strains, cracking pattern, and failure mode with high precision in relation to the experimental results. From the analyses presented, the main conclusions are:

- Finite elements with quadratic interpolation order and complete integration were more suitable than quadratic elements with reduced integration, and linear elements with reduced or complete integration. In addition, it is recommended to use at least six layers of quadratic finite elements in terms of slab's height to reproduce the punching shear failure mode;
- Incoherent values of the dilatancy angle and viscosity can modify the failure mode of the concrete, generating a structure behavior that is inconsistent with experimental results. The values of $\psi = 25^\circ$ and $\mu = 10^{-4}$ provided excellent results for the model analyzed in this work;
- The tensile strength of concrete is a crucial parameter in determining the load capacity of flat slabs subjected to punching shear in being able to increase the slab's stiffness, delaying the onset of cracking, and consequently, the failure of the material for small increments of its value;

- The higher the concrete's modulus of elasticity, the stiffer the structure becomes; however, it reduces its ductility and load capacity. On the other hand, a lower modulus of elasticity is related to greater deformability, ductility, and load capacity of the numerical slab.

In future research, it is recommended to evaluate the proposed numerical modeling with other experimental specimens that cover different design specifications, such as geometry, mechanical properties of concrete, flexural reinforcement arrangements, and the inclusion of shear reinforcement. Thus, parametric investigations of design variables related to the design of slabs subject to punching shear and verification of the calculation methods present in the technical codes will be feasible.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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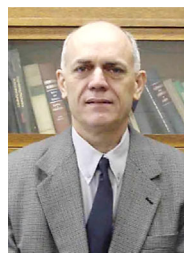
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